
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a central requirement for reliable machine
2 learning systems operating beyond closed-world training conditions. Recent work
3 has shown that unlabeled in-the-wild data can improve OOD detection, but this
4 setting remains technically difficult because the available unlabeled pool is itself a
5 mixture of in-distribution (InD) and OOD samples. Without a clean auxiliary OOD
6 set, the detector must learn from a contaminated source whose individual sample
7 identities are unknown. We introduce Medix, a median-centric framework that
8 extracts candidate outliers from unlabeled wild data and uses them, together with
9 labeled InD data, to train a binary OOD detector. Medix exploits the robustness of
10 the median as a stable estimate of central gradient behavior: samples whose removal
11 moves the element-wise median of wild-data gradients closer to the InD reference
12 gradient are treated as candidate OOD points. We provide error bounds showing
13 that the resulting filtering rule has controlled misclassification rates under the stated
14 contamination and concentration conditions. Across open-world benchmarks,
15 Medix consistently improves over existing baselines, supporting the theoretical
16 analysis and establishing median-based filtering as an effective mechanism for
17 learning OOD detectors from unlabeled wild mixtures.

1 Introduction

19 Machine learning systems deployed in realistic environments must operate under safety and reliability
20 requirements that are rarely captured by closed-world training data. A central failure mode arises
21 from out-of-distribution (OOD) inputs: samples originating from categories or sources outside the
22 training label space should not be assigned ordinary in-distribution (InD) predictions. Standard
23 neural networks, however, remain vulnerable to such inputs and often lack an explicit mechanism for
24 detecting and rejecting them at inference time (Nguyen et al., 2015).

25 OOD identification is difficult precisely because the relevant unknown distributions are absent from
26 supervised training. This limitation has motivated recent approaches that incorporate additional
27 “in-the-wild” data into OOD detector learning (Katz-Samuels et al., 2022a; Du et al., 2024a). In
28 particular, Katz-Samuels et al. (2022a) use unlabeled wild data to regularize training while preserving
29 the primary goal of classifying labeled InD examples. The attraction of this setting is practical as
30 well as statistical: unlabeled wild data can be collected after deployment from the environment in
31 which the model actually operates, thereby exposing the learning procedure to samples that better
32 resemble the OOD inputs encountered at test time.

33 The same practical advantage also creates the main technical obstacle. Wild data is not a clean OOD
34 resource; it is a contaminated mixture of InD and OOD samples, and the learner is not told which
35 component generated any individual point. Consequently, an OOD detector trained from this source
36 must first separate useful outlier evidence from the mixture, rather than merely consume a labeled or
37 clean auxiliary set. Despite its importance, this problem remains comparatively underdeveloped, and

theoretical understanding is especially limited: to the best of our knowledge, Du et al. (2024a) is the only prior work that establishes formal guarantees for the unlabeled “in-the-wild” setting. Meanwhile, recent studies have demonstrated the effectiveness of median-based approaches in data pruning (Acharya et al., 2024). Prompted by these developments, this paper aims to answer the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question by developing a median-centric perspective on OOD detection from unlabeled mixtures. The resulting framework, Medix, formulates outlier extraction as a median-based optimization problem over gradient representations and implements a greedy filtering algorithm that identifies candidate OOD samples with low error. The analysis shows that the fraction of InD samples incorrectly flagged as outliers is controlled by two interpretable effects: a contamination term, which captures the influence of OOD points and remains bounded when the OOD proportion is below 50%, and a concentration term, which uses the sub-Gaussian behavior of InD gradients to control their deviation from the mean InD gradient. Together, these terms provide high-probability guarantees for median-based filtering in the unlabeled in-the-wild setting and explain why the method remains robust even under adverse mixture configurations covered by the assumptions.

Empirically, we evaluate Medix against two families of baselines: methods trained only with InD data and methods that additionally use auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves the average FPR95 over the strong KNN+ baseline (Sun et al., 2022) by 40.98%. In contrast to Outlier Exposure (Hendrycks et al., 2019), which assumes a clean auxiliary dataset that is separable from InD data, Medix operates directly on the contaminated wild mixture and still achieves stronger performance. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, showing that explicit median-based filtering strengthens OOD detector training in the wild-data regime.

The paper makes three connected contributions:

- C1)** We introduce Medix, a median-centric framework whose core filtering stage extracts candidate outliers from unlabeled wild data and then trains an OOD detector using the extracted candidates together with labeled InD samples.
- C2)** We establish theoretical guarantees for median-based filtering in unlabeled mixtures. The analysis bounds both InD samples incorrectly flagged as outliers and OOD samples retained as inliers through the contamination effect, which is controlled when the OOD proportion is below 50%, and the concentration effect, which stabilizes the gradient behavior of InD samples.
- C3)** We evaluate Medix across eleven InD-OOD pairs against 20 competitive baselines. The results show consistent gains over all baselines, and the outlier-extraction study reports a low filtering error rate (e.g., 12.5%; see Figure 2), empirically corroborating the theoretical behavior of the median filter.

2 Preliminaries and Problem Setup

This section formalizes the OOD detection setting considered throughout the paper by specifying the labeled InD sample, the unlabeled wild mixture, the classifier and detector parameterizations, and the evaluation objectives.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and let $\mathcal{Y} = \{1, \dots, K\}$ denote the label space for InD classes. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ consists of n independently and identically distributed (i.i.d.) samples from $\mathbb{P}_{XY}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ and use $\mathcal{S}_{\text{in}}$ to train the InD model.

Out-of-distribution detection. We study the deployment scenario in which a model trained on labeled InD data encounters inputs whose labels are outside the training label space, i.e., for some label $y \notin \mathcal{Y}$. The detector should accept inputs generated by the InD distribution and reject inputs generated by an OOD source, thereby preventing ordinary class predictions on samples outside the modeled label space.

87 **Unlabeled wild data.** A persistent obstacle in OOD detection is the lack of labeled OOD samples:
 88 the space of possible OOD inputs is large, heterogeneous, and costly to annotate. Following Katz-
 89 Samuels et al. (2022a), we therefore incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into
 90 the learning framework. This wild sample is a deployment-relevant mixture of InD and OOD inputs
 91 that can be collected after training a model on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a); Katz-Samuels
 92 et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize the marginal
 93 distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

94 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The case
 95 $\pi = 0$ contains no OOD samples and therefore reduces to a trivial detection problem.

96 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameter $\phi \in \Phi$,
 97 where Φ is the classifier parameter space. Its output represents scores over the $K = |\mathcal{Y}|$ InD
 98 classes, and the labeled InD loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For detection, we introduce a binary
 99 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameter $\theta \in \Theta$. Its binary loss is denoted by $\ell_b(g_\theta(x), y_b)$, where
 100 $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, with $y_+ > 0$ assigned to InD samples and $y_- < 0$ assigned to OOD samples.

101 **Learning objective.** The learning objective uses both the labeled InD distribution $\mathbb{P}_{\text{in}}$ and the
 102 unlabeled wild distribution $\mathbb{P}_{\text{wild}}$ to train the OOD detector g_θ and the multi-class classifier f_ϕ . At
 103 test time, we evaluate detection and classification performance by the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

104 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

105 3 Method: Median-Centric Framework for OOD Detection

106 This section presents **Medix**, a median-centric learning framework for OOD detection with unlabeled
 107 wild data. Rather than relying solely on labeled InD samples or assuming a clean auxiliary OOD
 108 set, Medix first extracts candidate outliers from the contaminated wild mixture and then uses those
 109 candidates to train a detector. The framework has two stages: **1) Outlier Extraction**: a median-based
 110 filtering procedure that isolates candidate OOD samples from unlabeled wild data (Section 3.1); and
 111 **2) Detector Training**: binary detector learning with labeled InD samples and the extracted outlier
 112 candidates (Section 3.2). For the second stage, we follow the protocol introduced by Du et al. (2024a).
 113 This decomposition makes the central contribution explicit: robust filtering transforms an unlabeled
 114 InD–OOD mixture into training evidence for an OOD detector, thereby matching the open-world
 115 setting in which clean OOD labels are unavailable.

116 3.1 Extracting Candidate Outliers from the Wild Data

117 To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix uses an optimization-based criterion
 118 defined on model-parameter gradients. These gradients are computed from a classifier f_ϕ trained only
 119 on the labeled InD sample $\mathcal{S}_{\text{in}}$, so deviations in wild-data gradient behavior are measured relative to
 120 an InD reference rather than to labeled OOD data.

121 **Reference gradient estimation from InD data.** The first step is to estimate a reference gradient
 122 from the labeled InD data $\mathcal{S}_{\text{in}}$. We train f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

123 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we compute the mean InD gradient $\bar{\nabla}_{\text{in}}$ by
 124 averaging the loss gradients with respect to the model parameters over the labeled InD sample:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\mathbf{x}_i), y_i). \quad (3)$$

This vector $\bar{\nabla}_{\text{in}}$ serves as the reference against which the aggregate gradient behavior of wild-data subsets is compared.

Motivation. The filtering criterion is motivated by the observation that increasing OOD contamination in $\mathcal{S}_{\text{wild}}$ should move the median wild-gradient profile away from the average InD gradient $\bar{\nabla}_{\text{in}}$. We examine this behavior using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples disjoint from the training set used to obtain $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN samples and measure the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of wild-data gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Figure 1 shows a clear monotonic increase in this deviation as OOD samples are added. This empirical pattern motivates the Medix filtering rule: removing genuine OOD points should reduce the median-gradient deviation toward the InD reference, and the algorithm can stop once consecutive iterations no longer produce a meaningful decrease, as determined by the threshold ϵ .

Filtering potential outliers from unlabeled wild data. Motivated by Figure 1, we formulate outlier extraction as the following optimization problem over subsets of the wild sample, with the goal of identifying $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \left\{ \nabla \ell(\tilde{\mathbf{x}}_i) \right\}_{i \in \mathcal{S}}$.

The optimization seeks a subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ whose EWM gradient is closest to the InD reference gradient $\bar{\nabla}_{\text{in}}$. In light of Figure 1, the minimizing subset $\mathcal{S}_{\text{in}}^*$ is interpreted as the InD component of the wild mixture, and its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ provides the corresponding candidate OOD set.

Directly solving (4) is computationally prohibitive for large $\mathcal{S}_{\text{wild}}$. Medix therefore uses the greedy leave-one-out approximation in Algorithm 1, which iteratively removes the samples whose exclusion most reduces the median-gradient deviation from the InD reference.

At iteration t , the algorithm computes the EWM of gradients for the remaining wild subset $\mathcal{S}$ and measures its distance from $\bar{\nabla}_{\text{in}}$ as $d_t = \left\| \text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}} \right\|$. For each candidate sample $i \in \mathcal{S}$, Medix recomputes the EWM after removing that sample, $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$, and evaluates the induced decrease $\delta_i = d_t - \left\| \text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}} \right\|$. The k samples producing the largest decreases are added to the outlier set and removed from $\mathcal{S}$. The procedure terminates when the largest remaining decrease is below ϵ or the maximum number of iterations is reached. This stopping rule follows the monotonic trend in Figure 1: after the principal OOD mass has been removed, additional removals yield only small changes in L_2 distance and increasingly risk discarding InD samples.

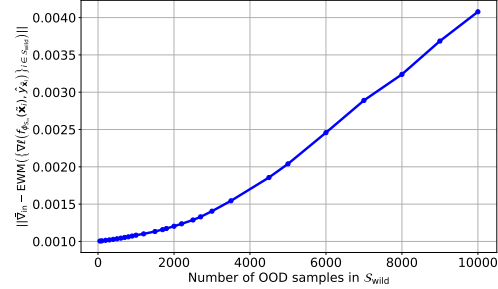


Figure 1: Median-gradient distance from the InD reference as OOD samples are added to $\mathcal{S}_{\text{wild}}$.

3.2 Training the OOD Detector with Candidate Outliers

After Medix identifies the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from wild data, we train the OOD detector g_{θ} to separate labeled InD samples from the extracted candidates, following the protocol of Du et al. (2024a). The objective enforces separability at the zero decision threshold by assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The resulting loss is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}, t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

Because the 0/1 loss is non-differentiable, we optimize a differentiable sigmoid-based binary surrogate that preserves the intended decision rule while making training tractable. The detector objective is combined with the InD risk in Equation 2, so detector training uses the extracted outlier signal without discarding the classifier’s InD predictive objective. This joint optimization supports the empirical generalization of g_θ to OOD samples drawn from $\mathbb{P}_{\text{out}}$.

4 Theoretical Analysis

We now formalize the behavior of Medix’s filtering stage. The following results upper-bound the two error modes induced by filtering: InD points incorrectly removed as outliers and OOD points incorrectly retained as inliers. These bounds make precise how EWM filtering depends on contamination, gradient concentration, and, for OOD retention, separation from the InD gradient profile. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 decomposes inlier filtering error into a *concentration term*, which decreases with sample size and accounts for finite-sample fluctuations of InD gradients, and a *contamination term*, which measures the influence of OOD gradients on the EWM. The bound remains controlled when the contamination ratio satisfies $\pi < 0.5$, matching the median’s robustness to minority contamination. The complementary question is how many true outliers are not removed; the next theorem bounds this OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in*

$\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

187

188 Theorem 4.2 complements the inlier result by bounding the fraction of OOD points mistakenly
 189 retained as InD. Together, Theorems 4.1 and 4.2 provide a two-sided robustness guarantee for Medix
 190 filtering. The two error rates are governed by three effects:

191 **Contamination Effect.** In Theorem 4.1, the median criterion remains stable under a nontrivial OOD
 192 fraction π provided $\pi < 0.5$. The term $\pi/[2(1 - \pi)]$ increases with π and captures the extent to
 193 which OOD gradients can shift the EWM. Conversely, Theorem 4.2 includes the reverse penalty $\frac{1-\pi}{2\pi}$,
 194 reflecting the difficulty of isolating OOD points when they are underrepresented in the mixture.

195 **Concentration Effect.** Both bounds use the sub-Gaussian behavior of gradient coordinates. For InD
 196 data, concentration near $\bar{\nabla}_{\text{in}}$ stabilizes the median under finite samples. For OOD data, concentration
 197 around μ_{out} quantifies the probability that an OOD gradient remains close enough to the InD profile
 198 to evade filtering. These terms decay at the stated $1/\sqrt{m}$ rate with the relevant sample size.

199 **Separation Effect.** Theorem 4.2 additionally requires separation between the OOD mean gradient and
 200 the InD reference gradient. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ expresses this dependence:
 201 larger separation makes it less likely that OOD gradients are retained by the median filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering controls both directions of error in unlabeled InD–OOD mixtures. InD removal is bounded by contamination and concentration effects, whereas OOD retention is controlled by separation, concentration, and reverse contamination. These results give Medix a principled basis for extracting outlier candidates before detector training.

202

203 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The coordinate-wise sub-Gaussian
 204 assumption on InD gradients enables the concentration bounds used in the analysis. Figure 4a
 205 provides empirical support for this assumption: the histogram of InD gradient values is concentrated
 206 around the mean and exhibits light tails, which is consistent with sub-Gaussian behavior. Figure 4b
 207 further compares empirical gradient quantiles with Gaussian theoretical quantiles in a Q-Q plot (Wilk
 208 and Gnanadesikan, 1968); the alignment with the 45-degree reference line indicates that the empirical
 209 gradient distribution is close to Gaussian, and Gaussian variables form a subclass of sub-Gaussian
 210 variables. We also provide a looser version of these bounds, which **removes the sub-Gaussian**
 211 **assumption** and holds under merely bounded second moments. This version is presented in Theorem
 212 C.3 (Appendix C.3). While the rates degrade, the core robustness guarantee of Medix still holds in
 213 this broader setting.

214 5 Experiments

215 This section will demonstrate the efficacy of Medix across various InD–OOD dataset pairs, bench-
 216 marking it against 20 widely-used baselines. All experiments are performed on hardware equipped
 217 with NVIDIA A100-SXM4-80GB GPUs. The necessary code to reproduce our results will be released
 218 upon acceptance.

219 5.1 Models, Datasets, and Baselines

220 **Datasets.** We use the same experimental protocol as Katz-Samuels et al. (2022a), which introduced
 221 the problem of learning OOD detectors with wild data, enabling a fair comparison to prior work.

Specifically, we use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we select a suite of natural image datasets, including PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing parameter $\pi = 0.5$. For example, when using PLACES365 as an OOD test set, we construct a wild mixture by combining CIFAR with PLACES365 as wild data and test on PLACES365 as the OOD set. This procedure is repeated across all OOD datasets and baselines. The InD CIFAR dataset is split into two halves: the first 25,000 images to train $\phi_{\mathcal{S}_{\text{in}}}$, while the remaining images to generate the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation, we use the penultimate layer weights, as these have been shown to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We evaluate using three standard metrics: (1) False Positive Rate (FPR95 $\downarrow$) of OOD samples when the True Positive Rate of InD samples is 95%, (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. Our comparison encompasses a diverse set of competitive baselines, categorized based on whether they are trained using only InD data or both InD and wild data. For the methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix against a variety of established OOD detection methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022); these methods use a model trained with softmax cross-entropy loss. Additionally, we also compare against methods based on contrastive loss, such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), for a more comprehensive comparison. For methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize the training by promoting lower confidence or higher energy on outlier data. We also include a comparison with WOODS (Katz-Samuels et al., 2022a), which introduced the concept of wild unlabeled data and utilizes it for OOD detection through a constrained optimization approach. Finally, we included more recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), to provide a more thorough evaluation.

5.2 Experimental Setup

In line with WOODS (Katz-Samuels et al., 2022a), we employ a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and a width factor of 2 for the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$. It is trained using stochastic gradient descent with a momentum of 0.9, weight decay of 0.0005, and an initial learning rate of 0.1. Training is performed for 100 epochs with cosine learning rate decay, a batch size of 128, and a dropout rate of 0.3. Hyperparameters ϵ and k used in the proposed method Medix are selected from the sets $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, taking into account dataset sizes and with the objective of maximizing OOD performance. For the OOD classifier g_{θ} , we initialize it with the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and add a linear layer that performs binary classification using the penultimate-layer features. The learning rate for this classifier is set to 0.001, and fine-tuning is done for 100 epochs as outlined in Equation (5). We combine the binary classification loss with the InD classification loss, assigning a weight of 10 to the binary classification component. All other training parameters remain the same as those used for training $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

We present our main results in Table 2 on CIFAR-100, where Medix remarkably outperforms all OOD detection baselines. The results highlight the following key observations: (1) Methods trained on both InD and wild data significantly outperform those trained exclusively on InD data. Medix reduces the FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES compared to KNN+, demonstrating the effectiveness of incorporating in-the-wild data for model regularization. (2) Medix further outperforms competitive methods utilizing wild data ($\mathbb{P}_{\text{wild}}$): On CIFAR-100, Medix achieves an average FPR95 of 5.42%, which represents a 1.32% improvement over WOODS. Additionally, Medix maintains a competitive InD accuracy of 73.33%. This slight difference can be attributed to the fact that our method is trained on 25,000 labeled InD samples, while baseline methods, which do not leverage wild data, use the full CIFAR-100 training set of 50,000 samples. We present the results for CIFAR-10 in Table 1, where Medix surpasses all baseline methods. Medix outperforms

Table 1: OOD detection performance comparison of Medix and baselines on CIFAR-10 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	89.57	81.44	90.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_m$ and $\mathbb{P}_{wid}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance comparison of Medix and baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_m$ only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
	Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in terms of FPR95, demonstrating its effectiveness in detecting OOD samples.

A representative visual example of Medix. We further investigate the performance of Algorithm 1 (Outlier Extraction) in extracting OOD samples from wild data $\mathcal{S}_{wild}$. To visualize this, we design an experiment using 2-dimensional synthetic data. This simulation is designed to be simple to facilitate better understanding. We generate the InD data by sampling from three multivariate Gaussian distributions, corresponding to three classes. The mean vectors are set to $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, respectively. The covariance matrix for all three classes is fixed at $0.25 \cdot I$. For the OOD data, we use a multivariate Gaussian distribution $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. In Figure 2, we observe that our method successfully identifies outliers, the majority of which align with the ground truth. Medix successfully flags 87.5% of actual OOD samples as outliers, underscoring its robustness in outlier extraction.

Additional studies and insights. Due to space constraints, we defer additional experiments and insights to Appendix A, which include (1) ablation studies on hyperparameter selection and sensitivity analysis (Appendix A.2), showing Medix’s strong robustness to hyperparameters, (2) a comparison between EWM and geometric median (Appendix A.1), showing that EWM is more sensitive to distributional shifts, making it more effective and reliable choice for filtering in our method, (3) a comprehensive comparison with methods employing competitive contrasting learning objectives (Appendix A.3), demonstrating that Medix outperforms even these competitive baselines by a significant margin, (4) evaluation on large-scale data under the complex unseen OOD setting, where $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), demonstrating that Medix outperforms baselines by a significant margin, (5) evaluating computation and memory efficiency of Medix (Appendix A.6) (6) evaluating the impact of pseudo-label quality (Appendix A.5), showing that our method is resilient to noisy or low-confidence labels, and (4) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

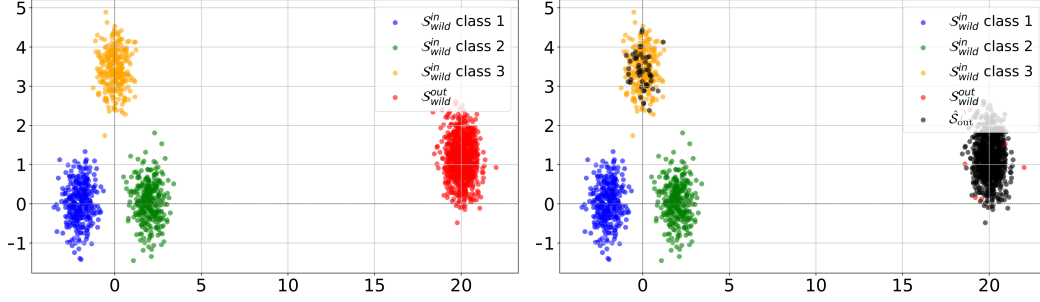


Figure 2: Example of Medix applied to unlabeled wild data. (a) Setup of the InD data $S_{\text{wild}}^{\text{in}}$ and OOD data $S_{\text{wild}}^{\text{out}}$ in the wild, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{S}_{\text{out}}$ filtered by Medix (in black), with an error rate of $\hat{S}_{\text{out}}$ containing InD data $S_{\text{wild}}^{\text{in}}$ is only 12.5%.

6 Related Work

In recent years, there has been a growing interest in OOD detection (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One approach to detect OOD data uses scoring functions to assess data distribution, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwal et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based score (Huang et al., 2021b), energy-based score (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

Another approach to OOD detection involves using regularization techniques during the training phase (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, regularization techniques can be applied to the model to either reduce its confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data. Most of these regularization methods assume the availability of an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) have relaxed the assumption of using only labeled data by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), though they did not propose a clear mechanism for outlier detection. In contrast, Du et al. (2024a,b) introduced an explicit outlier filtering method, but their thresholding technique differs fundamentally from ours, as we utilize a new median-centric approach to detect the outliers. Additionally, Katz-Samuels et al. (2022a); Du et al. (2024a) operate under the assumption of batch-level mixing, where each batch has a set ratio of InD and OOD samples. However, with large outsourced datasets, such structured mixing is not available—data is mixed randomly across the dataset. Our method addresses this by enabling dataset-level mixing without relying on batch-level structure. Many studies also leverage positive-unlabeled learning, which trains classifiers using positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). However, a key distinction from our approach is that these methods focus solely on differentiating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$, without simultaneously training an OOD classifier. Additionally, we propose a median-centric method to identify outliers in unlabeled data, with provably low error rates.

7 Conclusions

In this work, we introduced Medix, a novel median-centric framework for OOD detection that leverages unlabeled in-the-wild data. Using the inherent robustness of median operation, Medix

effectively filters outliers from mixed unlabeled data, enabling the training of a reliable OOD detector. Our theoretical analysis established provable bounds on the inlier misclassification rate, demonstrating that Medix maintains robustness even under significant OOD contamination (up to 50%), with errors controlled by sub-Gaussian concentration and contamination effects. We also provided complementary theoretical limits on the rate of OOD misclassification to accurately isolate OOD samples under clear separation conditions. Empirical validation across diverse benchmarks showcased Medix’s performance superiority over 20 baselines: it reduced the average FPR95 by 40.98% compared to strong baselines like KNN+ and outperformed state-of-the-art methods such as WOODS and DRL, while achieving an outlier extraction error rate as low as 12.5%.

References

- Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.
- Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning. *arXiv preprint arXiv:2406.17188*.
- Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.
- Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting wild data for both out-of-distribution generalization and detection. In *International Conference on Machine Learning*, pages 1454–1471. PMLR.
- Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1563–1572.
- Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.
- Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.
- Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-distribution detection. *arXiv preprint arXiv:2209.09858*.
- Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-distribution detection. In *The Eleventh International Conference on Learning Representations*.
- Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution detection? In *The Twelfth International Conference on Learning Representations*.
- Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard: Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.
- Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.
- Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.
- Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13678–13688.
- Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis. In *International Conference on Learning Representations*.
- Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 16068–16078.

385 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
386 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural*
387 *Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

388 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
389 *Advances in Neural Information Processing Systems*, 34:7068–7081.

390 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
391 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International*
392 *Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
393 1059, New York, New York, USA. PMLR.

394 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
395 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

396 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
397 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information*
398 *Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

399 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
400 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
401 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
402 Associates, Inc.

403 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
404 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine*
405 *Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

406 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
407 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

408 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
409 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

410 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI*
411 *Conference on Artificial Intelligence*, volume 32.

412 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
413 semi-supervised learning. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
414 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
415 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.

416 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
417 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
418 4858–4866.

419 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
420 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference*
421 *on computer vision and pattern recognition*, pages 41–50.

422 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
423 neural networks. *arXiv preprint arXiv:1610.02136*.

424 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
425 neural networks. In *International Conference on Learning Representations*.

426 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
427 *International Conference on Learning Representations*.

428 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
429 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings*
430 *of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.

431 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
432 wild. *Advances in Neural Information Processing Systems*, 34:677–689.

433 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
434 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information*
435 *Processing Systems*.

436 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
437 101.

438 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
439 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
440 *Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.

441 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
442 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.

443 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
444 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings*
445 *of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning*
446 *Research*, pages 10848–10865. PMLR.

447 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF*
448 *Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.

449 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
450 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.

451 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.

452 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
453 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
454 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
455 Inc.

456 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
457 distribution samples. *arXiv preprint arXiv:1711.09325*.

458 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
459 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

460 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
461 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
462 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
463 Inc.

464 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International*
465 *Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

466 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
467 neural networks. In *International Conference on Learning Representations*.

468 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
469 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
470 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
471 volume 33, pages 7498–7512. Curran Associates, Inc.

472 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in*
473 *neural information processing systems*, 33:21464–21475.

474 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
475 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing*
476 *Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

477 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
478 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and
479 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

480 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
481 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information*
482 *Processing Systems*, volume 31. Curran Associates, Inc.

483 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
484 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and Garnett,
485 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

486 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don’t know. In
487 *International Conference on Learning Representations*.

488 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
489 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th*
490 *International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
491 pages 15650–15665. PMLR.

492 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
493 models? *International Journal of Computer Vision*, 132(2):596–609.

494 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
495 detection? In *The Eleventh International Conference on Learning Representations*.

496 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
497 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
498 volume 2011, page 4. Granada.

499 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
500 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern*
501 *recognition*, pages 427–436.

502 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
503 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
504 29.

505 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
506 distribution detection. In *The Twelfth International Conference on Learning Representations*.

507 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
508 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
509 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

510 Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis
511 distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

512 Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution
513 detection and selective generation for conditional language models. In *The Eleventh International Conference*
514 *on Learning Representations*.

515 Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set
516 consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W.,
517 editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran
518 Associates, Inc.

519 Schwag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection.
520 In *International Conference on Learning Representations*.

521 Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in*
522 *Neural Information Processing Systems*, 34:144–157.

523 Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European*
524 *Conference on Computer Vision*, pages 691–708. Springer.

525 Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In
526 *International Conference on Machine Learning*, pages 20827–20840. PMLR.

527 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally
528 shifted instances. *Advances in neural information processing systems*, 33:11839–11852.

529 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally
530 shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
531 *Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.

532 Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International*
533 *Conference on Learning Representations*.

534 Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect
535 for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the*
536 *61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
537 12813–12832, Toronto, Canada. Association for Computational Linguistics.

538 Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep
539 deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.

540 Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In
541 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.

542 Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they
543 don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.

544 Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for
545 out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine,
546 S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran
547 Associates, Inc.

548 Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-
549 distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.

550 Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution
551 detection with implicit outlier transformation. In *The Eleventh International Conference on Learning*
552 *Representations*.

553 Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via
554 conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K.,
555 and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425.
556 Curran Associates, Inc.

557 Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scosmatch: Alleviating overtrusting
558 in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.

559 Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence
560 with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S.,
561 editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings*
562 *of Machine Learning Research*, pages 23631–23644. PMLR.

563 Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong
564 learning. In *International Conference on Learning Representations*.

565 Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*,
566 55(1):1–17.

567 Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural
568 networks. In *The Eleventh International Conference on Learning Representations*.

569 Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages
570 205–219. PMLR.

571 Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent
572 out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*,
573 pages 8301–8309.

574 Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al.
575 (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information*
576 *Processing Systems*, 35:32598–32611.

577 Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International*
578 *Journal of Computer Vision*, 132(12):5635–5662.

579 Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale
580 image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

581 Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

582 Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023).
583 Openood v1. 5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

584 Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space
585 for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing*
586 *Systems*.

587 Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label
588 distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
589 *Recognition*, pages 14461–14470.

590 Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection
591 learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*,
592 36:72110–72123.

593 Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for
594 scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

595 Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence
596 of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
597 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran
598 Associates, Inc.

599 Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure
600 for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing*
601 *Systems*, 36:22702–22734.

602 Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue', H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting
603 out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave,
604 D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages
605 20758–20769. Curran Associates, Inc.

606 A Additional Studies

607 A.1 Element-wise Median vs Geometric Median

608 Another pertinent question to ask is why we chose the element-wise median
 609 over the geometric median (Acharya et al., 2024) for outlier detection in Medix.
 610 To answer this question, we conducted a preliminary experiment using CIFAR-10 as the InD dataset and SVHN as the
 611 OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 5k samples
 612 drawn from CIFAR-10, ensuring that these samples are
 613 disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$,
 614 which we leverage to compute $\bar{\nabla}_{\text{in}}$. We incrementally add
 615 SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track the behavior of
 616 the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise
 617 median of the gradients of the wild dataset as well as the
 618 proportion of OOD samples removed. The results, shown
 619 in the Figure 3, demonstrate that for the same number of
 620 OOD samples in the wild dataset, the element-wise median
 621 identified a significantly higher proportion of OOD
 622 samples as outliers compared to the geometric median.
 623 This indicates that the element-wise median is more sensitive to distributional shifts, making it a
 624 more effective and reliable choice for filtering in our method.
 625

Table 3: Effect of hyperparameters ϵ and k on OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

626 A.2 Hyperparameter Selection and Sensitivity Analysis

627 We conducted an ablation study to assess the sensitivity of Medix to the hyperparameters ϵ and k .
 628 For this experiment, we employed CIFAR-100 as the InD dataset and SVHN as the OOD dataset.
 629 As shown in Table 3, Medix exhibits strong robustness to variations in the values of ϵ and k . In
 630 particular, Medix achieves a remarkably low FPR95 of 0.16 when using $\epsilon = 0.005$ and $k = 20000$.
 631 Even when the values of ϵ and k are varied—e.g., reducing ϵ to 0.0005 or halving k to 10000—the
 632 performance remains competitive (FPR95 of 0.20 and 0.68, respectively), demonstrating a graceful
 633 degradation rather than a sharp drop. The results indicate that while optimal hyperparameter selection
 634 is important, Medix maintains its effectiveness across a variety of hyperparameter choices, surpassing
 635 the baselines. This insensitivity to exact hyperparameter tuning makes Medix a reliable choice in
 636 real-world deployments, where exhaustive tuning may not be feasible.

637 A.3 Comprehensive comparison with recent competitive methods

638 We have extended our comparisons to include competitive baselines that employ diverse contrasting
 639 learning objectives, such as SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER
 640 Ming et al. (2023). Although these methods use contrasting learning objectives, which we do not,
 641 we included them for a comprehensive comparison. Additionally, we included more recent methods,
 642 such as CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du
 643 et al., 2022c), to provide a more thorough evaluation. For this experiment, we use CIFAR-100 as InD

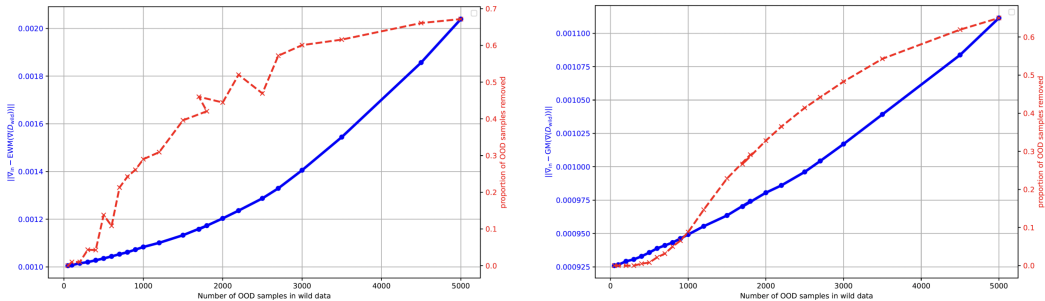


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM).

Table 4: OOD detection performance comparison of Medix and recent competitive baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

dataset and train the ResNet-34 model¹ following the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained using stochastic gradient descent with momentum 0.9, and weight decay 10^{-4} for 500 epochs. The initial learning rate is 0.5 with cosine scheduling and the batch size is 512. We use the checkpoints provided by Ming et al. (2023)². The results, as shown in Table 4, demonstrate that Medix outperforms even these competitive baselines by a significant margin, including those trained with contrastive learning objectives, achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Furthermore, Medix provides a provably low error rate, offering theoretical guarantees that most alternative approaches do not.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next investigate whether Medix can handle large-scale wild OOD data under the unseen OOD setting. This setting is more challenging for two main reasons: (1) it leverages a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally chosen to be distributionally different from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which better reflects realistic and challenging deployment scenarios. We used CIFAR-100 as the labeled InD data, 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data and tested on the SVHN dataset as the test OOD data. This setup introduces a greater level of complexity because the large-scale wild OOD data (300K Random Images) is significantly different from the OOD test data (SVHN). To evaluate the performance of our approach, we compare it against baselines that also leverage wild data in training. The results in the Table 5 highlight the superior performance of our method, Medix, compared to the baselines, with a significantly lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6), showing its effectiveness in distinguishing between InD and OOD data.

Table 5: Comparison of OOD detection performance on large-scale unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

Another important question to ask is whether the quality of pseudo-labels $\hat{y}_{\tilde{x}_i}$, particularly in terms of softmax confidence, affects the performance of OOD filtering, and whether a simple pre-filtering step that removes low-confidence pseudo-labels could improve the robustness of the model. To answer

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

this question, we conducted an experiment to evaluate how filtering low-confidence pseudo-labels affects OOD filtering. For this experiment, we used CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Specifically, we computed the softmax probabilities for each pseudo-labeled sample and discarded those with low-confidence predictions, setting a threshold of 0.6. After filtering, we found that 15.98% of the samples were removed from the training set.

The results showed that there was virtually no difference in performance between the method with filtering and the method without filtering. Both methods yielded a FPR95 of 0.01%, but with filtering, AUROC increased slightly from 99.98% to 99.99%. Thus, removing low-confidence pseudo-labels doesn't significantly impact the robustness of the model or its ability to detect OOD samples, showing that our method is resilient to noisy or low-confidence labels, and further filtering steps are unlikely to yield meaningful improvements.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now turn to the question of whether Medix's filtering phase is computationally feasible and memory-efficient on large datasets. In response to this question, we conducted profiling experiments on an NVIDIA A100-SXM4-80GB GPU using approximately 15,000 unlabeled samples for each InD–OOD pair. For this experiment, we used CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As seen from the results in Table 6, the GPU memory usage remains modest, and the filtering phase completes within a tractable timeframe, even when handling thousands of unlabeled samples. This demonstrates that the Medix filtering process is computationally feasible and does not impose excessive memory overhead, even on large datasets.

Table 6: Profiling results for Medix filtering on NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several key ways. For example, in OpenML Saito et al. (2021), the main problem is to handle outliers in semi-supervised learning (SSL) when training a standard classifier, whereas in our setting, the main challenge is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector classifier. In other words, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, our approach enhances OOD detection in an open-world setting where labeled OOD data is unavailable.

Secondly, these SSL methods aim to train a classifier that is robust to OOD samples in SSL, treating outliers as noise, while in our setting, the goal is to explicitly detect OOD samples and train an OOD detector. In other words, SSL methods focus on suppressing OOD samples during training to improve SSL performance, whereas Medix actively detects and leverages these OOD samples to build a more effective and reliable OOD detection system.

Lastly, the two approaches differ in their techniques: OpenMatch is based on consistency regularization, where the main idea is to enforce consistency across different stochastic transformations of the same input. This is mathematically modeled through soft constraints on the classifier's outputs, encouraging smooth decision boundaries. On the other hand, Medix relies on median-based filtering for outlier detection. The method uses statistical robustness properties of the median, leveraging its insensitivity to extreme values to identify potential OOD samples. The theoretical analysis derives error bounds based on the contamination effect (proportion of OOD samples) and the concentration effect (sub-Gaussian behavior of InD gradients), ensuring that OOD detection error remains low.

718 B Broader Impacts and Limitations

719 As machine learning continues to advance, tackling the challenges of OOD detection has become
720 essential for ensuring robust and reliable model performance in real-world applications. We introduce
721 a median-centric framework for OOD detection that enhances OOD handling and model safety.
722 The impact of our research goes beyond theoretical advancements, with practical applications in
723 healthcare, autonomous systems, and finance. By improving OOD detection, we address a key
724 challenge in model deployment, fostering greater trust and adoption of machine learning technologies.
725 Future work could explore integrating it with generative models for outlier synthesis or adapting it to
726 dynamic environments where OOD distributions evolve over time.

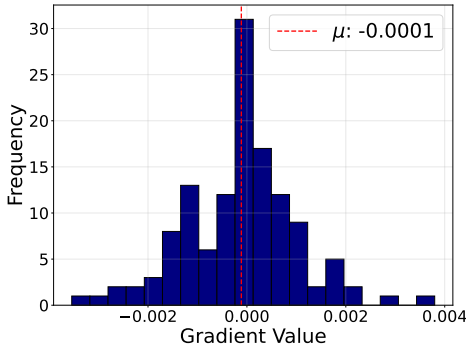
727 A potential limitation of Medix lies in its reliance on a mixture of unlabeled InD and OOD data,
728 which, in practice, may be noisy, corrupted, or inconsistently sampled. However, this assumption is
729 not unique to our method—it reflects the inherent uncertainty of real-world deployment environments
730 and is a common starting point in robust learning theory. To make our assumptions explicit and
731 verifiable, we rigorously formalize them in the main theorems and provide a precise geometric
732 condition in Remark 4.3 to justify the necessary separation required for reliable filtering. Moreover,
733 we extend our guarantees to looser settings in Theorem C.3, removing the need for sub-Gaussian tails
734 and demonstrating the resilience of our framework under weaker distributional assumptions. These
735 efforts underscore that while Medix does make simplifying assumptions, they are both interpretable
736 and relaxable, forming a principled foundation for future improvements in OOD detection under
737 realistic noise conditions.

738 C Proof of Theorem

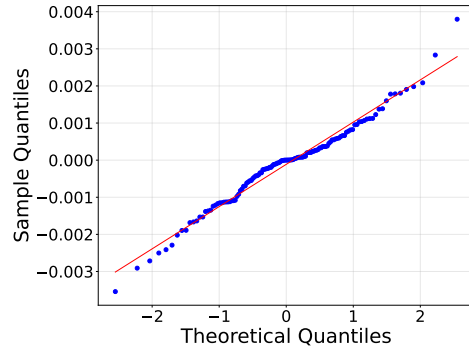
739 To aid clarity and consistency throughout the theoretical and empirical sections, we summarize the
 740 key notation used in this paper in Table 7. This includes definitions for gradient-related quantities,
 741 sample sizes, and filtering terms, as well as parameters that appear in our concentration bounds.
 742 Unless stated otherwise, we use bold symbols for vectors, $\mathcal{X}$ to denote the input space, and assume all
 743 gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

744 C.1 Bound on the Inlier Misclassification Rate

745 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 746 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 747 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 748 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

749 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
 750 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\nabla}_{\text{in}}$. We define
 751 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
 752 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
 753 than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
 754 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict
 755 optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification
 756 rate, ERR_{in} .

757 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 758 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 759 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

760 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

761 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

762 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

763 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

764 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 765 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

766 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

767 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 768 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

769 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

770 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 771 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

772 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

773 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

774 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
775 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

776 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

777 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

778 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

779 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

780 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
781 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
782 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
783 called *good inliers*.

784 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 785 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

786 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 787 thereby contradicting the optimality of $\mathcal{S}^*$.

788 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 789 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

790 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

791 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

792 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

793 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 794 available for swapping in.

795 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 796 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

797 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

798 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

799 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

800 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

801 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 802 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

803 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

804 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

805 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

806 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

807 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

808 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
809 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

810 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
811 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
812 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

813 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

814 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
815 increase it, so the median distance from μ_j does not increase and possibly decreases.

816 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

817 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

818 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
819 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

820 □

821 C.2 Bound on the Outlier Retention Rate

822 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
823 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
824 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
825 between the OOD and InD gradient means.

826 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
827 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

828 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

829 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

830 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

831 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

832 *Proof. Step 1: Concentration of OOD Gradients.*

833 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
834 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

835 has probability at most $p_{\text{out}}(\varepsilon)$.

836 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

837 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

838 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

839 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

840 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

841 which holds with probability at least $1 - \delta$.

842 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

843 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

864 **Differences with the original bound.** The original bound, which assumes sub-Gaussianity, features
 865 a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

866 that decreases exponentially with dimension d , making the bound extremely tight for large d . In
 867 contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay
 868 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

869 that decays as $O(1/d)$, yielding a looser bound for larger d . The original bound requires sub-Gaussian
 870 tails, while the new one only assumes finite fourth moments, allowing for heavier tails. In particular,
 871 both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

872 that shrinks as m_{in} increases, independent of the tail assumptions.

873 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 874 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 875 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 876 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 877 bound derived from the finite fourth moment assumption.

878 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

879 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 880 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

881 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 882 S_{wild} .

883 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 884 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

885 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

886 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

887 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

888 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

889 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

890 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

891 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

892 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
893 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

894 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
895 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

896 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

897 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

898 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

899 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 900 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

901 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 902 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

903 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 904 thereby contradicting the optimality of $\mathcal{S}^*$.

905 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 906 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

907 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

908 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

909 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

910 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 911 available for swapping in.

912 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 913 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

914 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

915 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

916 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

917 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

918 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 919 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

920 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

921 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

922 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

923 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

924 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

925 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 926 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

927 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 928 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 929 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

930 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

931 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 932 increase it, so the median distance from μ_j does not increase and possibly decreases.

933 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

934 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

935 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 936 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

937 □